Customer Transaction Prediction

This project aims to build a machine learning model that predicts whether a customer will make a future transaction based on anonymized banking data.

📌 Project Overview

Banks and financial institutions use transaction prediction to identify potential customers likely to engage in future activity. Predicting transactions helps improve marketing campaigns, customer retention strategies, and personalized offers.

In this project, we:

- Explored anonymized customer banking data.
- Processed and cleaned the dataset for training.
- Applied machine learning models to predict transaction likelihood.
- Evaluated model performance using accuracy, precision, recall, and F1-score.

📂 Dataset

The dataset contains anonymized customer information, including:

- **Customer demographics** (age, tenure, etc.)
- **Banking activity features** (account balance, product holdings, etc.)
- **Transaction history** (binary labels: whether a transaction occurred or not).

Note: Dataset is anonymized to protect customer privacy.

⚙️ Tech Stack

- **Programming Language**: Python
- **Libraries/Frameworks**:
    - pandas, numpy → Data preprocessing
    - matplotlib, seaborn → Data visualization
    - scikit-learn → Model building and evaluation
    - xgboost, lightgbm → Advanced predictive models

🚀 Approach

1. **Data Preprocessing**

    - Handling missing values
    - Feature scaling & encoding
    - Train-test split
2. **Exploratory Data Analysis (EDA)**
    - Distribution of transactions
    - Customer behavior patterns
    - Correlation between features
3. **Modeling**
    - Logistic Regression
    - Random Forest
    - Gradient Boosting (XGBoost, LightGBM)
4. **Evaluation Metrics**
    - Accuracy
    - Precision, Recall, F1-score
    - ROC-AUC

📊 Results

- Gradient Boosting models outperformed baseline classifiers.
- Achieved high ROC-AUC score, making it reliable for identifying potential transaction customers.

🔮 Future Work

- Hyperparameter tuning for better performance.
- Deploying the model as an API.
- Real-time prediction system integration.

📜 License

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